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Bivariate-Multivariate Hybrid Models For Purchase Intent Prediction

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Introduction

- With the rapid growth of e-commerce, predicting purchase intent has become essential for improving recommendations, marketing strategies, and conversion rates in purchases.
- Traditional ML models capture mainly linear or bivariate relationships and struggle with complex feature interactions.
- This work proposes a bivariate–multivariate hybrid framework for purchase intent prediction using clickstream data.

Objective

- To investigate the role of bivariate and multivariate feature interactions in modeling consumer purchase behavior from clickstream data.
- To design and implement bivariate interaction model using Factorization Machines for purchase intent prediction.
- To model complex non-linear and higher-order feature dependencies using deep-learning based multivariate approaches.
- To develop and evaluate bivariate–multivariate hybrid models, including DeepFM, xDeepFM, and AutoInt, for improved predictive performance.
- To perform a comparative analysis of traditional, bivariate, multivariate, and hybrid models using standard evaluation metrics on imbalanced e-commerce data.

Methodolgy

- Pipeline: Data preprocessing → imbalance handling → model training → evaluation
- Dataset: Real-world e-commerce clickstream data (session-level), binary target (purchase / no purchase), highly imbalanced
- Preprocessing:

1. Label encoding for categorical features
2. Standardization of numerical features
3. Train–test split

- Class Imbalance Handling: SMOTE applied to training data to oversample purchase sessions
- Modeling Approaches:

1. Bivariate: Factorization Machines for pairwise feature interactions
2. Multivariate: Deep neural network for nonlinear, higher-order patterns
3. Hybrid:

- DeepFM (FM + DNN)
- xDeepFM (explicit interaction modeling)
- AutoInt (self-attention–based interactions)
- Training Setup: PyTorch implementation, Adam optimizer, binary cross-entropy loss

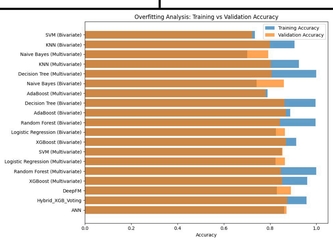
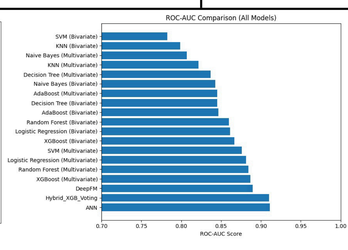
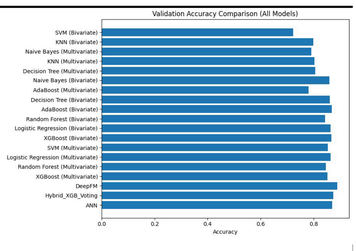
- Evaluation Metrics: Accuracy, Precision, Recall, F1-score, AUC (ROC)

Results

- FM improved performance by modeling pairwise feature interactions but showed lower recall for purchase sessions.
- DNN captured nonlinear behavioral patterns, resulting in better overall performance than FM alone.
- Hybrid models achieved the highest F1-score and AUC by combining low-order and high-order feature interactions.
- Applying SMOTE increased recall and AUC across all models, improving detection of rare purchase events.

Applications

- Personalized Recommendation
- Targeted Digital Marketing
- E-commerce Conversion Optimization
- Customer Behavior Analysis
- Real-Time Decision Support Systems
- Business Intelligence & Analytics

Conclusion

- By integrating bivariate interaction modeling through FM with multivariate representation learning using deep and attention-based models, the proposed approach effectively captures both direct feature relationships and complex consumer behavior patterns.
- The results highlight the importance of modeling feature interactions at different levels of complexity for accurate purchase intent prediction. The proposed hybrid framework offers a robust and scalable solution for understanding consumer behavior and can be effectively applied to recommendation systems and targeted marketing.